

CSCI E-106:Assignment 8

Due Date: April 10, 2026 at 11:59 pm EST

Instructions

Students should submit their reports on Canvas. The report needs to clearly state what question is being solved, step-by-step walk-through solutions, and final answers clearly indicated. Please solve by hand where appropriate.

Please submit two files: (1) a R Markdown file (.Rmd extension) and (2) a PDF document, word, or html generated using knitr for the .Rmd file submitted in (1) where appropriate. Please, use RStudio Cloud for your solutions.

Problem 1

Using the diabetes data under faraway library,if you are using the R Studio, please use the command below to download the data:

```
library(faraway)
data(diabetes,package="faraway")
str(diabetes)
```

```
## 'data.frame':    403 obs. of  19 variables:
## $ id      : 'labelled' int  1000 1001 1002 1003 1005 1008 1011 1015 1016 1022 ...
##  ..- attr(*, "label")= chr "Subject ID"
## $ chol     : 'labelled' int  203 165 228 78 249 248 195 227 177 263 ...
##  ..- attr(*, "label")= chr "Total Cholesterol"
## $ stab.glu : 'labelled' int  82 97 92 93 90 94 92 75 87 89 ...
##  ..- attr(*, "label")= chr "Stabilized Glucose"
## $ hdl      : 'labelled' int  56 24 37 12 28 69 41 44 49 40 ...
##  ..- attr(*, "label")= chr "High Density Lipoprotein"
## $ ratio    : 'labelled' num  3.6 6.9 6.2 6.5 8.9 ...
##  ..- attr(*, "label")= chr "Cholesterol/HDL Ratio"
## $ glyhb    : 'labelled' num  4.31 4.44 4.64 4.63 7.72 ...
##  ..- attr(*, "label")= chr "Glycosolated Hemoglobin"
## $ location: Factor w/ 2 levels "Buckingham","Louisa": 1 1 1 1 1 1 1 1 1 1 ...
## $ age      : int    46 29 58 67 64 34 30 37 45 55 ...
##  ..- attr(*, "units")= chr "years"
## $ gender   : Factor w/ 2 levels "male","female": 2 2 2 1 1 1 1 1 1 2 ...
## $ height   : int    62 64 61 67 68 71 69 59 69 63 ...
##  ..- attr(*, "units")= chr "inches"
## $ weight   : int   121 218 256 119 183 190 191 170 166 202 ...
##  ..- attr(*, "units")= chr "pounds"
## $ frame    : Factor w/ 3 levels "small","medium",...: 2 3 3 3 2 3 2 2 3 1 ...
```

```
## $ bp.1s : 'labelled' int 118 112 190 110 138 132 161 NA 160 108 ...
## .. attr(*, "label")= chr "First Systolic Blood Pressure"
## $ bp.1d : 'labelled' int 59 68 92 50 80 86 112 NA 80 72 ...
## .. attr(*, "label")= chr "First Diastolic Blood Pressure"
## $ bp.2s : 'labelled' int NA NA 185 NA NA NA 161 NA 128 NA ...
## .. attr(*, "label")= chr "Second Systolic Blood Pressure"
## .. attr(*, "comment")= chr "equals first measurement if it was not high"
## $ bp.2d : 'labelled' int NA NA 92 NA NA NA 112 NA 86 NA ...
## .. attr(*, "comment")= chr "equals first measurement if it was not high"
## .. attr(*, "label")= chr "Second Diastolic Blood Pressure"
## $ waist : int 29 46 49 33 44 36 46 34 34 45 ...
## .. attr(*, "units")= chr "inches"
## $ hip : int 38 48 57 38 41 42 49 39 40 50 ...
## .. attr(*, "units")= chr "inches"
## $ time.ppn: 'labelled' int 720 360 180 480 300 195 720 1020 300 240 ...
## .. attr(*, "label")= chr "Postprandial Time when Labs were Drawn"
## .. attr(*, "units")= chr "minutes"
```

If you are using Posit R Cloud instead of RStudio Desktop, please use the command below to download the data:

```
tmp <- tempfile(fileext = ".rda")
download.file(
  "https://raw.githubusercontent.com/cran/faraway/master/data/diabetes.rda",
  destfile = tmp,
  mode = "wb"
)
load(tmp)
head(diabetes)
```

```
##      id chol stab.glu hdl ratio glyhb location age gender height weight frame
## 1 1000 203      82 56 3.6 4.31 Buckingham 46 female 62 121 medium
## 2 1001 165      97 24 6.9 4.44 Buckingham 29 female 64 218 large
## 3 1002 228      92 37 6.2 4.64 Buckingham 58 female 61 256 large
## 4 1003 78      93 12 6.5 4.63 Buckingham 67 male 67 119 large
## 5 1005 249      90 28 8.9 7.72 Buckingham 64 male 68 183 medium
## 6 1008 248      94 69 3.6 4.81 Buckingham 34 male 71 190 large
## bp.1s bp.1d bp.2s bp.2d waist hip time.ppn
## 1 118 59 NA NA 29 38 720
## 2 112 68 NA NA 46 48 360
## 3 190 92 185 92 49 57 180
## 4 110 50 NA NA 33 38 480
## 5 138 80 NA NA 44 41 300
## 6 132 86 NA NA 36 42 195
```

Four hundred three African Americans were interviewed in a study to understand the prevalence of obesity, diabetes and other cardiovascular risk factors in central Virginia. In this question we build a regression tree-based model for predicting glycosolated hemoglobin (glyhb) in terms of the other relevant variables. (70 points)

a-) Plot the response against each of the predictors and comment on the apparent strength of the relationships observed. (5 pts)

b-) Investigate the pattern of missing values in the data. By eliminating a combination of rows and columns, produce a reduced dataset that contains no missing values. Try to make the reduction as small as possible. (5 pts)

c-) Fit a linear model with main effects for all the predictors. Which predictors appear to be most important? Make a plot of the residuals and fitted values from this model and comment. (10 pts)

d-) Fit the regression tree. From the output, answer the following questions: What is the mean response for these observations? What characterizes the largest terminal node in the tree? What is the mean response in this node? (10 pts)

e-) Make a plot of the tree. What feature of the plot reveals the most important predictor? (10 pts)

f-) Plot the residuals against the fitted values for this tree. Comment. (10 pts)

g-) Predict the response for an individual with these predictor values: (10 pts)

```
dat<-data.frame(cbind(id=1004,chol=213,stab.glu=72,hdl=58,ratio=3.3,location="Buckingham",
age=56,gender="female",height=64,weight=131,frame="medium",bp.1s=108,
bp.1d=55,bp.2s=NA,bp.2d=NA,waist=30,hip=40,time.ppn=720))
```

(h) Glycosylated hemoglobin greater than 7.0 is usually taken as a positive diagnosis of diabetes. Build the classification tree for the diagnosis of diabetes and compare this model to the corresponding regression tree. Which predictors are most useful? (10 pts)

Problem 2

Refer to the Prostate cancer data set. Serum prostate-specific antigen (PSA) was determined in 97 men with advanced prostate cancer. PSA is a well-established screening test for prostate cancer and the oncologists wanted to examine the correlation between level of PSA and a number of clinical measures for men who were about to undergo radical prostatectomy. The measures are cancer volume, prostate weight, patient age, the amount of benign prostatic hyperplasia, seminal vesicle invasion, capsular penetration, and Gleason score. (30 pts)

a-) Select a random sample of 65 observations to use as the model-building data set (Please use `set.seed(567)`). Develop a best subset model for predicting PSA on the train data set. Justify your choice of model. Assess your model's ability to predict and discuss its usefulness to the oncologists. (5 pts)

b-) Develop a regression tree for predicting PSA on the train data set. Justify your choice of number of regions (tree size), and interpret your regression tree. (10 pts)

c-) Compare the performance of your regression tree model with that of the best regression model on the test and train data sets. Which model would you choose and why? (10 pts)

d-) Which model is more easily interpreted and why? (5 pts)